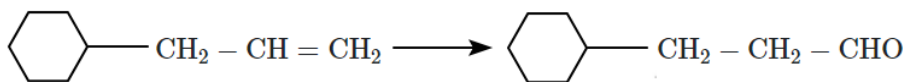


Q1. Identify the correct reagents that would bring about the following transformation.

NEET - 2024



1. (i) BH_3 (i) BH_3
 (ii) $\text{H}_2\text{O}_2/\ominus\text{OH}$ (ii) $\text{H}_2\text{O}_2/\text{OH}^\ominus$
 (iii) PCC (iii) PCC
2. (i) BH_3 (i) BH_3
 (ii) $\text{H}_2\text{O}_2/\ominus\text{OH}$ (ii) $\text{H}_2\text{O}_2/\text{OH}^\ominus$
 (iii) alk. KMnO_4 (iii) alk. KMnO_4
 (iv) H_3O^+ (iv) H_3O^+
3. (i) $\text{H}_2\text{O}/\text{H}^+$ (i) $\text{H}_2\text{O}/\text{H}^+$
 (ii) PCC (ii) PCC
4. (i) $\text{H}_2\text{O}/\text{H}^+$ (i) $\text{H}_2\text{O}/\text{H}^+$
 (ii) CrO

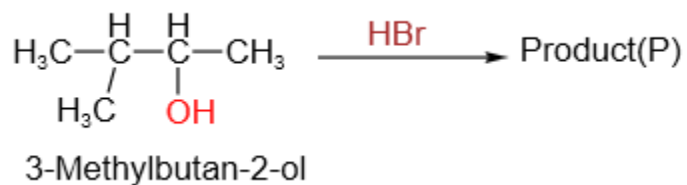
Q2. Which one of the following alcohols reacts instantaneously with Lucas reagent?

NEET - 2024

1.	$\begin{array}{c} \text{CH}_3 - \text{CH}_2 - \text{CH} - \text{OH} \\ \\ \text{CH}_3 \end{array}$
2.	$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2\text{OH} \\ \\ \text{CH}_3 \end{array}$

3. propanal
4. propionic acid

Q5. Consider the following reaction and identify the product (P). **NEET - 2023**



1.	$ \begin{array}{c} \text{H}_3\text{C} \quad \text{H}_2 \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array} $	2.	$ \begin{array}{c} \text{Br} \quad \text{H}_2 \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} $
3.	$\text{H}_3\text{CHC}=\text{CHCH}_3$	4.	$ \begin{array}{c} \text{Br} \\ \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array} $

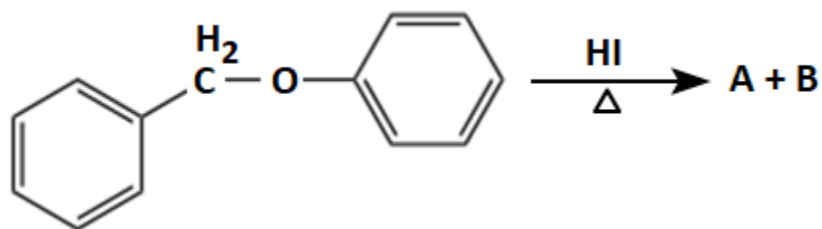
Q6. Which amongst the following will be most readily dehydrated under acidic conditions? **NEET - 2023**

1.	$ \begin{array}{c} \text{NO}_2 \\ \\ \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH} \end{array} $	2.	$ \begin{array}{c} \text{NO}_2 \quad \text{OH} \\ \quad \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{H} \end{array} $
3.	$ \begin{array}{c} \text{OH} \quad \text{OH} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \\ \text{H} \end{array} $	4.	$ \begin{array}{c} \text{NO}_2 \\ \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{CH}_2-\text{OH} \\ \quad \\ \text{H} \quad \text{OH} \end{array} $



Q7. Consider the following reaction:

NEET - 2023



Identify products A and B.

1.	$\text{A} = \text{C}_6\text{H}_5\text{CH}_3$ and $\text{B} = \text{C}_6\text{H}_5\text{I}$
2.	$\text{A} = \text{C}_6\text{H}_5\text{CH}_3$ and $\text{B} = \text{C}_6\text{H}_5\text{OH}$
3.	$\text{A} = \text{C}_6\text{H}_5\text{CH}_2\text{OH}$ and $\text{B} = \text{C}_6\text{H}_5\text{I}$
4.	$\text{A} = \text{C}_6\text{H}_5\text{CH}_2\text{I}$ and $\text{B} = \text{C}_6\text{H}_5\text{OH}$

Q8. Reagents that can be used to convert alcohols to carboxylic acids are : NEET - 2023

(A) $\text{CrO}_3 - \text{H}_2\text{SO}_4$

(B) $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4$

(C) $\text{KMnO}_4 + \text{KOH}/\text{H}_3\text{O}^+$

(D) $\text{Cu}, 573 \text{ K}$

(E) $\text{CrO}_3 + (\text{CH}_3\text{CO})_2\text{O}$

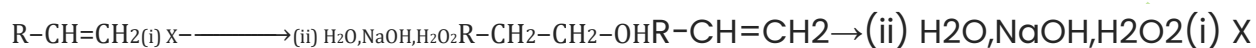
Choose the most appropriate answer from the options given below:

1. (B), (C) and (D) only	2. (B), (D) and (E) only
3. (A), (B) and (C) only	4. (A), (B) and (E) only



Q9. Consider the following reactions:

NEET - 2023

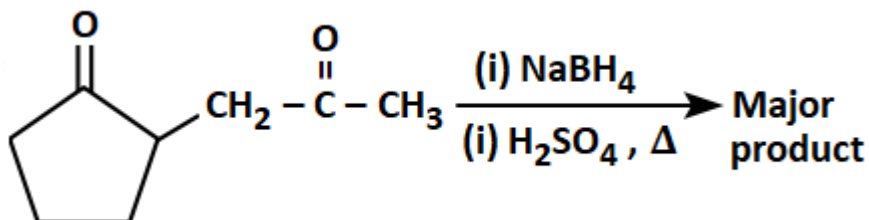


Identify 'X' in above reactions:

1.	B_2H_6	2.	$LiAlH_4$
3.	$NaBH_4$	4.	H_2/Pd

Q10. The major product formed in the following conversion is:

NEET - 2023



1.		2.	
3.		4.	

Q11. Given below are two statements:

NEET - 2022

I:	The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.
II:	o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.



NEET - 2022



NEET - 2022

In light
given b

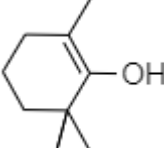
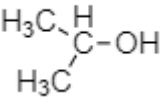
NEET - 2022

In light
given b

In light
given b

3.	Both I and II are correct
4.	Both I and II are incorrect

Q14. The increasing order of reactivity of the following compounds towards acid-catalyzed dehydration is: NEET - 2022

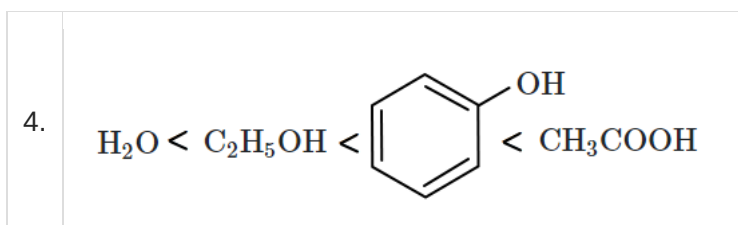
1.	$\text{CH}_3\text{-CH}_2\text{-OHCH}_3\text{-CH}_2\text{-OH}$	2.	
3.		4.	$\text{CH}_3\text{-C(CH}_3)_2\text{-OHCH}_3\text{-C(CH}_3)_2\text{-OH}$

1.	(II) < (III) < (I) < (IV)	2.	(II) < (I) < (III) < (IV)
3.	(I) < (III) < (IV) < (II)	4.	(III) < (I) < (II) < (IV)

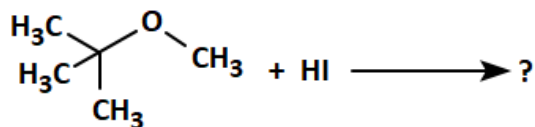
Q15. The correct order of acidic strength of the following molecules is: NEET - 2022

1.	$\text{H}_2\text{O} < \text{C}_6\text{H}_5\text{OH} < \text{C}_2\text{H}_5\text{OH} < \text{CH}_3\text{COOH}$
2.	$\text{C}_2\text{H}_5\text{OH} < \text{H}_2\text{O} < \text{CH}_3\text{COOH} < \text{C}_6\text{H}_5\text{OH}$
3.	$\text{C}_2\text{H}_5\text{OH} < \text{H}_2\text{O} < \text{C}_6\text{H}_5\text{OH} < \text{CH}_3\text{COOH}$





Q16. The major products formed in the following reaction are: NEET – 2022



1.	$(\text{CH}_3)_3\text{COH}, \text{CH}_3\text{I}$	2.	$(\text{CH}_3)_2\text{C}=\text{CH}_2, \text{CH}_3\text{I}$
3.	$(\text{CH}_3)_3\text{CI}, \text{CH}_3\text{OH}$	4.	$(\text{CH}_3)_3\text{COH}, \text{CH}_3\text{OH}$

Q17. Match the reagents (List-I) with the products (List-II) obtained from phenol:

NEET – 2022

	List-I		List-II
(A)	(i) NaOH (ii) CO_2 (iii) H^+	(I)	Benzoquinone
(B)	(i) Aqueous NaOH + CHCl_3 (ii) H^+	(II)	Benzene
(C)	Zn dust, $\Delta\Delta$	(III)	Salicylaldehyde
(D)	$\text{Na}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4$	(IV)	Salicylic acid

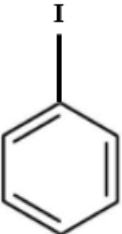
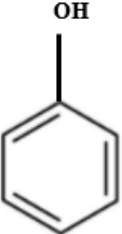
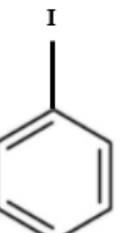
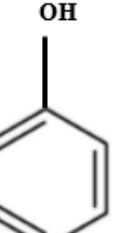
Choose the correct answer from the options given below:



1. Pentan-3-ol	2. 2-Methyl butan-2-ol
3. 2-Methyl propan-2-ol	4. Pentan-2-ol

Q20. Anisole on reaction with HI gives:

NEET - 2021

1.	 + CH_3COOH	2.	 + $\text{C}_2\text{H}_5\text{I}$
3.	 + $\text{C}_2\text{H}_5\text{OH}$	4.	 + CH_3I

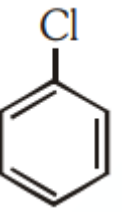
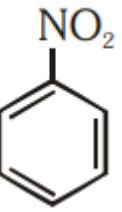
Q21. A reaction between acetone and methylmagnesium chloride followed by hydrolysis will give:

NEET - 2020

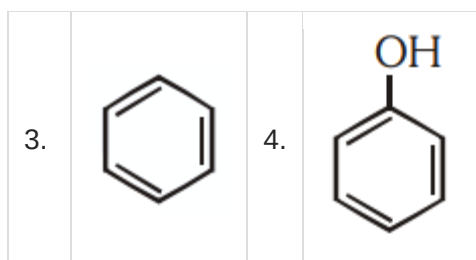
1. sec-Butyl alcohol	2. tert-Butyl alcohol
3. iso-Butyl alcohol	4. iso-Propyl alcohol

Q22. The most reactive compound among the following for electrophilic aromatic substitution is:

NEET - 2020

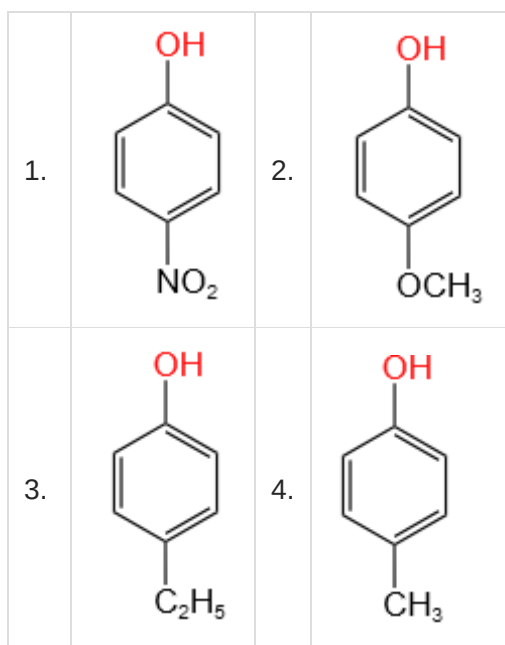
1.		2.	
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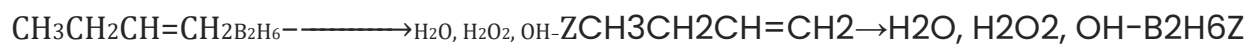
Q23. The most acidic compound among the following is:

NEET - 2020



Q24. Identify Z in the following reaction:

NEET - 2020

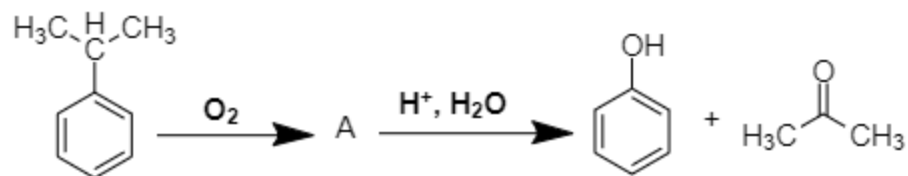


1.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	2.	$\text{CH}_3\text{CH}_2\text{C}(\text{H})(\text{CH}_3)\text{CH}_2\text{OH}$ $\text{CH}_3\text{CH}_2\text{C}(\text{H})(\text{CH}_3)\text{CH}_2\text{OH}$
3.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	4.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$



Q25. The structure of intermediate A in the following reaction is:

NEET - 2019



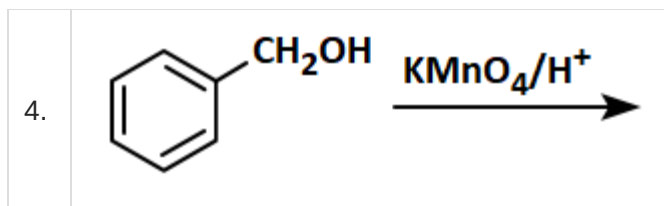
1.		2.	
3.		4.	

Q26. Mark the reaction among the following that does not give benzoic acid as the major product:

NEET - 2019

1.	
2.	
3.	



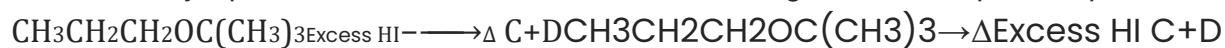


Q27. The vapor of secondary alcohol passed over heated copper at 573 K yield:

NEET – 2019

1. A carboxylic acid	2. An aldehyde
3. A ketone	4. An alkene

Q28. The major products C and D formed in the following reaction respectively are:



NEET – 2019

1. $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{I}$ and $\text{I}-\text{C}(\text{CH}_3)_3$
2. $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$ and $\text{I}-\text{C}(\text{CH}_3)_3$
3. $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{I}$ and $\text{HO}-\text{C}(\text{CH}_3)_3$
4. $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{OH}$ and $\text{HO}-\text{C}(\text{CH}_3)_3$



ANSWER KEY

1. Ans. (1)
2. Ans. (3)
3. Ans. (3)
4. Ans. (1)
5. Ans. (2)
6. Ans. (3)
7. Ans. (4)
8. Ans. (3)
9. Ans. (1)
10. Ans. (1)
11. Ans. (4)
12. Ans. (4)
13. Ans. (3)
14. Ans. (2)
15. Ans. (3)
16. Ans. (3)
17. Ans. (3)
18. Ans. (3)
19. Ans. (2)
20. Ans. (4)
21. Ans. (2)
22. Ans. (4)
23. Ans. (1)
24. Ans. (1)
25. Ans. (3)
26. Ans. (3)
27. Ans. (3)
28. Ans. (1)

